



TRAINING EXAMPLE — NOT A REAL IEC 62304 DELIVERABLE

This project is a **training site, not a real medical device** under any regulatory definition. The device described in this document — “SentinelFlow 500” — is entirely **fictional**, invented for this course: nothing here is a genuine IEC 62304 deliverable, a real regulatory submission, or evidence of any real organisation’s compliance with anything. It illustrates the *kind* of document a real SaMD/SiMD project would produce. See the [document register](#) for the full explanation.

Clause: 5.4 — Software Detailed Design · **Register:** [Device Example Register](#)

Document ID: SOP-05 · **Device:** SentinelFlow 500 (fictional) · **Safety class:** C

Status: SOP — template (Procedure not yet written for SentinelFlow 500)

Fictional worked example. See [SOP-01](#) for the disclaimer that applies to this whole document set, and the [Device Example Register](#) for the SentinelFlow 500 device description and what “template” status means here.

1. Purpose

This SOP defines how each software item in SentinelFlow 500's architecture is subdivided into software units and designed in enough detail to implement correctly, per Clause 5.4.

2. Scope

Applies to detailed design of SentinelFlow 500's control software (Class C — see [SOP-01 §7.3](#)). Covers Clause 5.4 in full. Note the split within the clause itself: subdividing into software units (§5.4.1) applies at Class B and C; the detailed design content, interface design, and design verification (§5.4.2–5.4.4) apply at Class C only — which is why this document, unlike most in this set, has

requirements that would not all apply if SentinelFlow 500 were reclassified to Class B.

3. Responsibilities

ROLE	RESPONSIBILITY UNDER THIS SOP
Software Architect	Subdivides each software item into software units (§5.4.1)
Software Development Lead	Owens the detailed design for each unit and its interfaces
QA/RA	Confirms the detailed design review record meets §5.4.4

4. Definitions & Abbreviations

TERM	MEANING
Software unit	A software item that is not subdivided further — the level at which detailed design and unit verification (Clause 5.5) operate
Detailed design	Design documented in enough detail that implementation requires only coding decisions, not further design decisions

5. References

- IEC 62304:2006+AMD1:2015, §5.4
- **SOP-04 — Software Architecture:** the software items this SOP subdivides into units
- **SOP-06 — Unit Implementation and Verification:** consumes each unit's detailed design as its implementation input

6. What the standard requires

REF	TITLE	APPLIES AT CLASS C	WHAT MUST BE PRODUCED
5.4.1	Subdivide software into software units	Yes	(No standard-named output — see the site's Learn page for why some rows carry no output entry.)
5.4.2	Develop detailed design for each software unit	Yes	Design documented in enough detail to allow correct implementation of each software unit
5.4.3	Develop detailed design for interfaces	Yes	Design documented for the interfaces of each software unit, in enough detail to implement them correctly
5.4.4	Verify detailed design	Yes	Documented verification that the detailed design implements the architecture and is free from contradiction with it

(Quoted directly from `applicability.json`, identical in content to the self-referential set's [05](#), whose Status is itself marked **Partial** — detailed design is written down for only one of that project's six scripts. This device-example set's version is a template for the same clause, not a completed alternative.)

7. Procedure

Not yet written for SentinelFlow 500 — see the [Device Example Register](#) for what "template" status means. Once complete, this section would give the detailed design for SentinelFlow 500's software units — e.g. the occlusion-detection algorithm, the dose-rate validation logic, the motor control state machine — down to the level a developer could implement without further design decisions.

5.4.1 — Subdivide software into software units

[Template.]

5.4.2–5.4.3 — Detailed design of units and their interfaces

[Template.]

5.4.4 — Verify detailed design

[Template.]

8. Records

RECORD	PRODUCED BY	RETAINED IN
Detailed design document per software unit	Software Development Lead	Project technical file
Detailed design review record	QA/RA	Project technical file

9. Revision History

Illustrative only — SentinelFlow 500 has no real revision history.

VERSION	STATUS	DESCRIPTION
1.0	Template	Structure and requirements table complete; Procedure section pending